

Scalability & Future Plan

Date	03 July 2026
Team ID	N/A
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how OptiCrop can be scaled to handle more users and data, and the roadmap for future enhancements.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Flask's built-in development server handles requests sequentially (single worker)	Limits how many concurrent users can get predictions smoothly	High
2	Training dataset is a representative synthetic dataset rather than large-scale real-world sensor data	May not fully capture regional/real-world variability in soil and climate conditions	Medium
3	No persistent database — model and dataset are file-based (CSV/Pickle)	Harder to track prediction history or support multi-user accounts as usage grows	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-threaded Flask dev server, local use only	Deploy behind Gunicorn/uWSGI with multiple workers, fronted by Nginx or a managed platform (Render/Heroku/AWS)
2	Data Storage	Static CSV dataset + Pickle model/scaler files	Move to a lightweight database (e.g., PostgreSQL/SQLite) for prediction logs and support larger, continuously updated datasets
3	Performance	Fast inference (~0.08s) but limited by sequential request handling	Add caching for repeated inputs and horizontal scaling (multiple app instances) as traffic grows
4	Security	No authentication; public, stateless prediction endpoint	Add basic rate-limiting and input validation hardening; introduce optional accounts if prediction history/personalization is added later

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate a live weather API for real-time rainfall/temperature input instead of manual entry	3–6 months	More accurate, up-to-date recommendations without requiring users to know exact readings
Phase 3	Add a researcher/policymaker dashboard aggregating prediction trends by region	6–9 months	Supports regional agricultural planning and policy decisions with real usage data
Phase 4	Deploy on cloud hosting (e.g., Render/AWS) with a production WSGI server and basic database	9–12 months	Enables real multi-user access, prediction history, and better reliability at scale